MINING DEALER BEST PRACTICE

OHT Brake Group Baseline Wear Indicator Plate

Maintenance and Repair | Application | **Component Life Management** | Component Renewal (CRC) | MARC Management

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1.0 Introduction

The Caterpillar wet oil-cooled, multiple disc brakes combine service, secondary, park brake and retarding functions into the same sealed robust system. This fully integrated system provides superior brake efficiency and gives operators the security they need to maintain high production levels.

In order to extend the braking system life, Caterpillar has released the availability of brake wear indicators for many current and legacy off-highway truck models. The brake wear indicator is used to quickly check brake disc wear. The service procedure is non-invasive, resulting in no brake fluid loss and the elimination of the need to bleed the brakes for routine service checks. This provides reduced risk to potential brake system contamination.

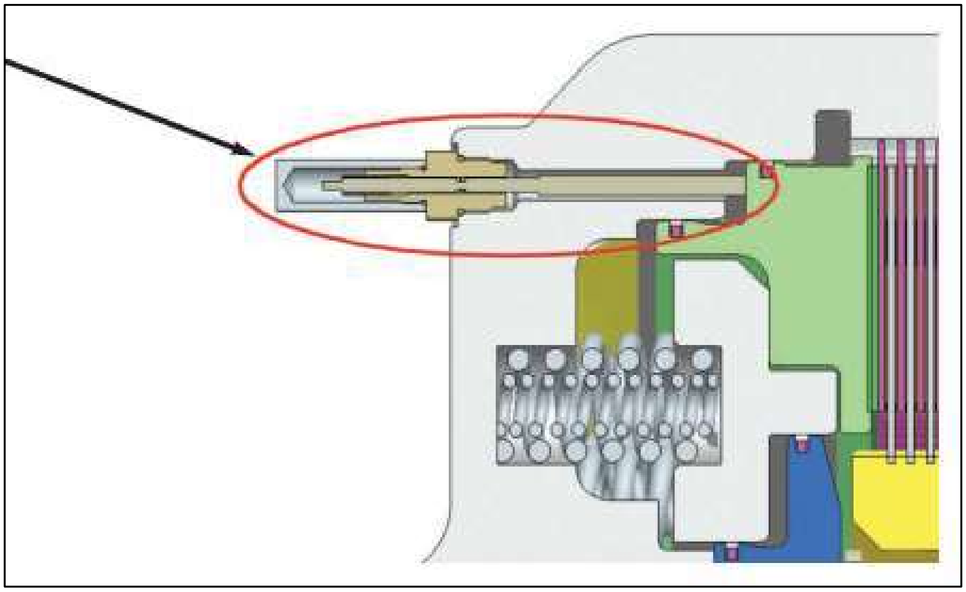

Figure 1 – Brake wear indicator for multiple disc brakes.

2.0 Best Practice Description

The brake wear indicator must be preset at the initial delivery and also be set again after any of the brakes are rebuilt. Performing the procedure after rebuilding the brakes creates a benchmark for the piston travel.

The procedure to establish a baseline is done by pushing the dowel bushing (D) into the body until the end of dowel bushing (D) is flush with the notched shoulder that is on the indicator rod (B). See the picture below for clarification. The full procedure to set the brake wear adjuster, is available in Sis 2.0; refer to Testing and Adjusting, "Installing and Setting the Brake Wear Indicator".

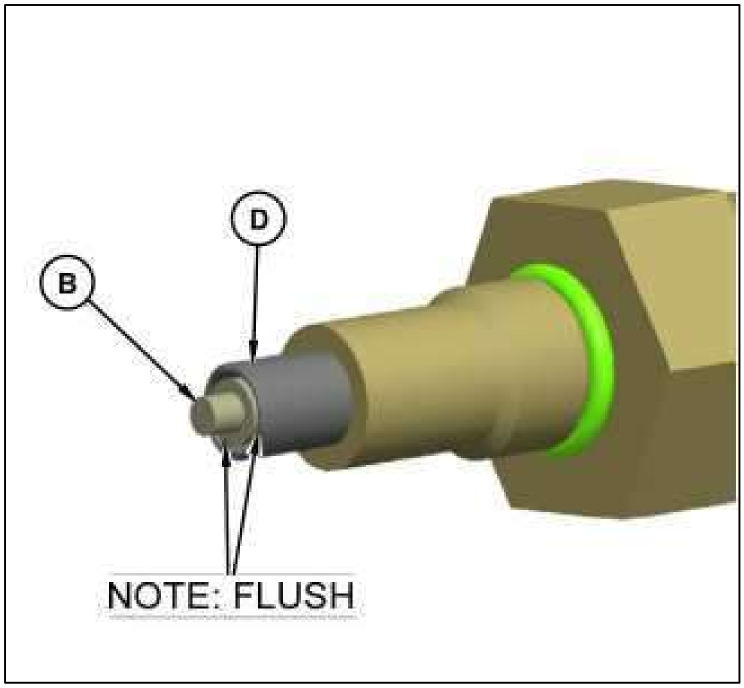

Figure 2 – Presetting the brake wear indicator after brakes reconditioning.

The best practice objective is to create a permanent record where the field personnel have prompt access to the component information and brake wear indicator baseline. The main information contained in the permanent tag is listed below:

- Component location
- Dowel bushing height
- Indicator rod height
- Work order number
- Assembly date

Figure 3 – Baseline plate installed on brake anchor.

3.0 Implementation Steps

New machine delivery:

1. Design a tag with the following information (dimensions 70 x 70mm)
    - Component location
    - Dowel bushing height
    - Indicator rod height
    - Work order number
    - Assembly date

2. Determine a common location to the brake groups, near the brake wear indicator

3. Before applying the new adhesive plate, ensure that the surface is clean and free of dirt, oil, grease, or other contaminants

4. Set the break wear indicator following the recommended procedure

5. Mark all of the information required on the aluminum tag

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777G Brake Gp Baseline	
Component Location	
Dowel Bushing Height	
Indicator Rod Height	
Work Order No.	
Assembly Date	

Figure 4 – Aluminum tag layout example, dimensions 70 x 70mm.

Component rebuild:

1. Design a tag with the following information (dimensions 70 x 70mm)
    - Component location
    - Dowel bushing height
    - Indicator rod height
    - Work order number
    - Assembly date

2. Determine a common location to the brake groups, near the brake wear indicator and the previous brake wear indicator plate

3. Before applying the new adhesive plate, ensure that the surface is clean and free of dirt, oil, grease, or other contaminants

6. After the component rebuild, set the break wear indicator following the recommended procedure

7. Mark all of the information required on the aluminum tag

4.0 Benefits

- Low-cost and easy to implement;
- Traceability of component data;
- Accurate brake wear measurement information;
- Improves brake life.

5.0 Resources Required

Fabricated aluminum tags.

6.0 Supporting Attachments / References

N/A.

7.0 Related Best Practices

N/A.

8.0 Acknowledgements

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